

Lab 1: Requirements Engineering & UML Use-Case Modelling

Problem Statement #45 — Multi-Cloud Asset Cost Optimization Portal

1. Requirements Table

Actors: *FinOps Analyst, Cloud Administrator*

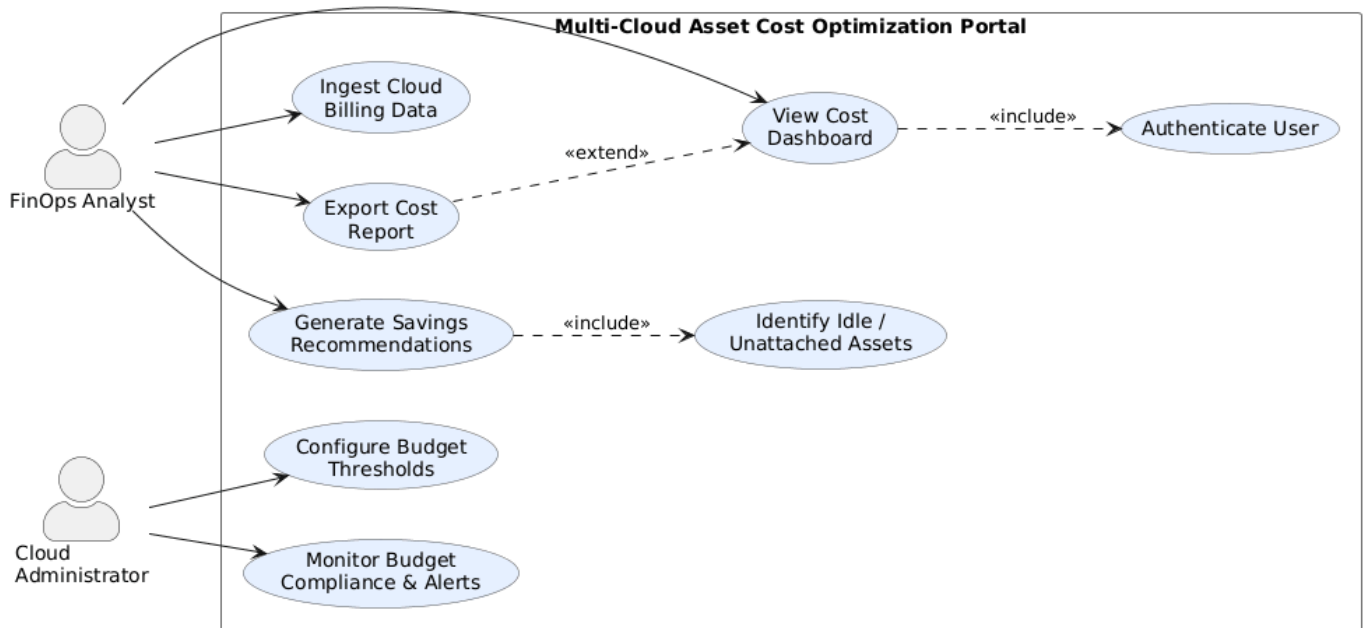
ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall ingest cloud billing CSV exports from AWS, Azure, and GCP, categorize costs by service and project tag, and highlight compute instances with < 5% CPU utilization over 14 days.	High	Pass: Idle cloud asset flagged with monthly savings estimate. Fail: Active instance misclassified as idle.	Cost visibility across providers is the foundation of FinOps; without accurate categorization and idle detection, no downstream optimization is possible.
FR-002	Functional	The system shall scan ingested billing/inventory data to detect unattached (orphaned) storage volumes and snapshots across all three cloud providers, and estimate their monthly cost impact.	High	Pass: An unattached volume older than 24 hours is listed with provider, region, size, and \$/month cost. Fail: A volume currently attached to a running instance is flagged as unattached.	Unattached storage is one of the most common sources of avoidable cloud spend and is explicitly called out in the problem context.
FR-003	Functional	The system shall allow a Cloud Administrator to define budget thresholds per project, tag, or cloud account, and shall automatically dispatch a breach alert (email/Slack webhook) when actual or forecasted spend exceeds the configured threshold.	High	Pass: Alert is dispatched within 5 minutes of a threshold breach being detected. Fail: No alert is sent, or an alert is sent when spend is below threshold.	Budget breach alerting is a core stated capability of the portal and enables proactive cost control rather than after-the-fact reporting.
FR-004	Functional	The system shall generate a consolidated cost-savings recommendation report (idle instances + unattached volumes + rightsizing suggestions) that a FinOps Analyst can export as CSV or PDF.	Medium	Pass: Exported report contains every asset flagged in the current analysis run with its estimated savings. Fail: Exported file omits flagged assets or shows stale data from a previous run.	Analysts need a shareable artifact to present savings opportunities to finance/engineering stakeholders outside the portal.
FR-005	Functional	The system shall support role-based authentication and authorization, restricting budget-threshold configuration to Cloud Administrators while allowing FinOps Analysts read/analysis access to all dashboards.	Medium	Pass: A FinOps Analyst attempting to change a budget threshold is denied with an authorization error. Fail: A FinOps Analyst is able to modify budget configuration.	Segregation of duties between spend-analysis and spend-governance roles is standard FinOps practice and prevents unauthorized configuration changes.

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Security	The portal must parse multi-gigabyte cloud billing files and update cost dashboards in under 2 minutes, while encrypting billing data in transit (TLS 1.2+) and at rest.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load. Fail: Dashboard refresh exceeds 2 minutes or data is transmitted/stored unencrypted.	Billing files can run into GBs for large enterprises, and billing data is commercially sensitive, so both speed and security are non-negotiable for adoption.
NFR-002	Reliability & Availability	The system shall maintain 99.5% monthly uptime and retain at least 13 months of historical billing/idle-asset data for trend analysis and audit purposes.	Medium	Pass: Uptime monitoring logs confirm $\geq 99.5\%$ availability over a calendar month; historical queries return data up to 13 months back. Fail: Unplanned downtime drops availability below 99.5%, or data older than 13 months is requested but unavailable and no graceful message is shown.	FinOps decisions rely on trend data (e.g., month-over-month spend), and budget alerting is only useful if the platform is reliably available when breaches occur.

2. UML Use-Case Diagram

The diagram below models both actors, all eight primary use cases, and includes one «include» relationship (View Cost Dashboard → Authenticate User; Generate Savings Recommendations → Identify Idle/Unattached Assets) and one «extend» relationship (Export Cost Report extends View Cost Dashboard).

Multi-Cloud Asset Cost Optimization Portal — Use-Case Diagram



3. Use-Case Flow Specification

Use Case: Identify Idle / Unattached Cloud Assets

Use Case ID	UC-03
Primary Actor	FinOps Analyst
Secondary Actor	Cloud Administrator (receives escalation on high-value findings)
Related Requirements	FR-001, FR-002
Trigger	FinOps Analyst requests an asset-optimization scan, or a scheduled nightly scan runs automatically after billing data ingestion.

Preconditions

1. The FinOps Analyst is authenticated and has an active session (via Authenticate User, included use case).
2. At least one billing/inventory CSV export from AWS, Azure, or GCP has already been ingested into the system (Ingest Cloud Billing Data) and successfully parsed.
3. The portal has resource-utilization metrics (e.g., CPU %, attachment status) available for the ingested period.

Postconditions

- Success: All idle compute instances (< 5% CPU over 14 days) and unattached storage volumes are flagged, each with an estimated monthly savings value, and the results are persisted for the current scan run.
- Failure: No assets are flagged, and the analyst is shown a clear error/empty-state message describing why (e.g., insufficient data, malformed file).

Main Success Scenario

1. The FinOps Analyst navigates to the Cost Dashboard and selects “Run Idle/Unattached Asset Scan.”
2. The system retrieves the most recently ingested billing and utilization data across AWS, Azure, and GCP.
3. The system evaluates each compute instance's average CPU utilization over the trailing 14 days against the < 5% idle threshold.
4. The system cross-references storage volumes/snapshots against active attachment records to identify unattached storage across all three providers.
5. The system calculates an estimated monthly savings figure for each flagged asset based on its provider list price and size/instance type.
6. The system displays the flagged assets in a sortable results table (provider, resource ID, region, type, idle reason, estimated savings).
7. The FinOps Analyst reviews the results and may forward high-impact findings to the Cloud Administrator, or proceed to Generate Savings Recommendations / Export Cost Report.

Alternate Flow — A1: Malformed or Incomplete Billing Data

Trigger: At Step 2, the system detects that the most recently ingested file is corrupted, missing required fields (e.g., resource ID, utilization metrics), or contains data for an incomplete period.

1. The system halts the scan and does not proceed to Steps 3–6.
2. The system displays an error message identifying the affected provider and the specific missing/invalid fields.
3. The system logs the failure and suggests the analyst re-ingest a corrected billing file (returns to Ingest Cloud Billing Data).
4. Use case ends in a failure postcondition; no assets are flagged for that provider, though results from unaffected providers (if any) are still shown.

Alternate Flow — A2: No Idle or Unattached Assets Found

Trigger: At Step 6, the scan completes but no assets meet the idle/unattached criteria.

1. The system displays an empty-state message (“No idle or unattached assets detected for this period”) rather than a blank table.
2. The FinOps Analyst may adjust the analysis window or re-run the scan later.

Notes

- This use case includes Authenticate User (every dashboard action requires a valid session).
- This use case is included by Generate Savings Recommendations, which consumes its flagged-asset output as input to the consolidated savings report.